

Homogeneous Linear Partial Differential Equations with Constant Coefficients

4.1 Homogeneous and Non-homogeneous linear equations with constant coefficients. A partial differential equation in which the dependent variable and its derivatives appear only in the first degree and are not multiplied together, their coefficients being constants or functions of x and y , is known as a *linear partial differential equation*. The general form of such an equation is

$$\left(A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n \frac{\partial^n z}{\partial y^n} \right) + \left(B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + B_1 \frac{\partial^{n-1} z}{\partial x^{n-2} \partial y} + \dots + B_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}} \right) + \left(M_0 \frac{\partial z}{\partial x} + M_1 \frac{\partial z}{\partial y} \right) + N_0 z = f(x, y), \dots (1)$$

where the coefficients $A_0, A_1, \dots, A_n, B_0, B_1, \dots, B_{n-1}, M_0, M_1$ and N_0 are constants or functions of x and y . If $A_0, A_1, \dots, A_n, B_0, B_1, \dots, B_{n-1}, M_0, M_1$ and N_0 are all constants, then (1) is called a *linear partial differential equation with constant coefficients*.

For convenience $\partial/\partial x$ and $\partial/\partial y$ will be denoted by D (or D_x) and D' (or D_y) respectively. Then (1) can be rewritten as

$$[(A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D'^n) + (B_0 D^{n-1} + B_1 D^{n-2} D' + \dots + B_{n-1} D'^{n-1}) + (M_0 D + M_1 D') + N_0]z = f(x, y), \dots (2)$$

or, briefly, $F(D, D')z = f(x, y)$(3)

When all the derivatives appearing in (1) are of the same order, then the resulting equation is called a *linear homogeneous partial differential equation with constant coefficients* and it is then of the form

$$(A_0 D^{n-1} + A_1 D^{n-1} D' + \dots + A_n D'^n)z = f(x, y). \dots (4)$$

On the other hand, when all the derivatives in (1) are not of the same order, then it is called a *non-homogeneous linear partial differential equation with constant coefficients*.

In this chapter we propose to study the various methods of solving homogeneous linear partial differential equation with constant coefficients, namely, (4)

4.2 Solution of a homogeneous linear partial differential equation with constant coefficients, namely, $(A_0 D^{n-1} + A_1 D^{n-1} D' + \dots + A_n D'^n)z = f(x, y)$, ...(1)

where $A_0, A_1, \dots, A_n$ are constants. (1) may be rewritten as

$$F(D, D')z = f(x, y), \dots (2)$$

where $F(D, D') = A_0 D^{n-1} + A_1 D^{n-1} D' + \dots + A_n D'^n$(3)

As in the case of linear ordinary differential equation with constant coefficients, we start with the following basic theorems.

Theorem I. If u is the complementary function and z' a particular integral of a linear partial differential equation $F(D, D')z = f(x, y)$, then $u + z'$ is a general solution of the equation.

Proof. Given $F(D, D')z = f(x, y)$(1)

The complementary function u of (1) is the most general solution of

$$F(D, D')z = 0. \dots (2)$$

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$$\therefore F(D, D')u = 0. \quad \dots(3)$$

Note that the complementary function must contain as many arbitrary constants as is the order of equation (2).

Any solution z' of (1) is called a particular integral of (1). Note that particular integral does not contain any arbitrary constant. Thus, by definition, we have

$$F(D, D')z' = f(x, y). \quad \dots(4)$$

Adding (3) and (4),

$$F(D, D')(u + z') = f(x, y),$$

showing that $u + z'$ is a solution of (1). Since (1) and (2) are of the same order, the general solution $u + z'$ will contain as many arbitrary constants as the general solution of (1) requires.

Theorem II. If $u_1, u_2, \dots, u_n$ are solutions of the homogeneous linear partial differential equation $F(D, D')z = 0$, then $\sum_{r=1}^n c_r u_r$ is also a solution, where $C_1, C_2, \dots, C_r, \dots, C_n$ are arbitrary constants.

$$\text{Proof. Given equation is } F(D, D')z = 0. \quad \dots(1)$$

$$\text{We have } F(D, D')(c_r u_r) = c_r F(D, D')u_r \quad \dots(2)$$

and

$$F(D, D')\sum_{r=1}^n v_r = \sum_{r=1}^n F(D, D')v_r \quad \dots(3)$$

for any set of functions v_r . Using results (2) and (3), we get

$$F(D, D')\sum_{r=1}^n (c_r u_r) = \sum_{r=1}^n F(D, D')(c_r u_r) = c_r \sum_{r=1}^n F(D, D')u_r \quad \dots(4)$$

Since u_r is solution of (1) for $r = 1, 2, \dots, n$, so $F(D, D')u_r = 0$ for $r = 1, 2, 3, \dots, n$.

$$\therefore (4) \text{ gives } F(D, D')\sum_{r=1}^n (c_r u_r) = 0, \text{ which proves the required result.}$$

Note. For convenience we shall denote complementary function by C.F. and particular integral by P.I.

4.3 Method of finding the complementary function (C.F.) of the linear homogeneous partial differential equation with constant coefficients, namely, $F(D, D')z = f(x, y)$

$$\text{i.e., } (A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D^n)z = f(x, y), \quad \dots(1)$$

where $A_0, A_1, \dots, A_n$ are all constants.

The complementary function of (1) is the general solution of

$$(A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D^n)z = 0. \quad \dots(2)$$

or

$$[(D - m_1 D')(D - m_2 D') \dots (D - m_n D')]z = 0. \quad \dots(3)$$

where $m_1, m_2, \dots, m_n$ are some constants.

Clearly, the solution of any one of the equations

$$(D - m_1 D')z = 0, \quad (D - m_2 D')z = 0, \dots, \quad (D - m_n D')z = 0 \quad \dots(4)$$

is also a solution of (3).

We now show that the general solution of $(D - mD')z = 0$ is $z = \phi(y + mx)$, where ϕ is an arbitrary function.

We have, $(D - mD')z = 0$ or $(\partial z / \partial x) - m(\partial z / \partial y) = 0$ or $p - mq = 0, \dots(5)$ which is in Lagrange's form $Pp + Qq = R$. Here Lagrange's auxiliary equations for (5) are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{0}. \quad \dots(6)$$

Taking the first two fractions of (6), $dy + m dx = 0$ so that $y + mx = c_1$... (7)

From the third fraction of (6), $dz = 0$ so that $z = c_2$ (8)

Hence from (7) and (8), the general solution of (5) is $z = \phi(y + mx)$, where ϕ is an arbitrary function. So, we assume that a solution of (2) is of the form

$$z = \phi(y + mx). \quad \dots(9)$$

$$\begin{aligned} \text{From (9),} \quad Dz &= \partial z / \partial x = m \phi'(y + mx), \\ D^2 z &= \partial^2 z / \partial x^2 = m^2 \phi''(y + mx), \end{aligned}$$

$$\text{and} \quad D^n z = \partial^n z / \partial x^n = m^n \phi^{(n)}(y + mx).$$

$$\begin{aligned} \text{Again,} \quad D' z &= \partial z / \partial y = \phi'(y + mx), \\ D'^2 z &= \partial^2 z / \partial y^2 = \phi''(y + mx), \end{aligned}$$

$$\text{and} \quad D'^n z = \partial^n z / \partial y^n = \phi^{(n)}(y + mx).$$

$$\text{Also, in general,} \quad D^r D'^s z = \partial^{r+s} z / \partial x^r \partial y^s = m^r \phi^{(r+s)}(y + mx).$$

Substituting these values in (2) and simplifying, we get

$$(A_0 m^n + A_1 m^{n-1} + A_2 m^{n-2} + \dots + A_n) \phi^{(n)}(y + mx) = 0,$$

which is true if m is a root of the equation

$$A_0 m^n + A_1 m^{n-1} + A_2 m^{n-2} + \dots + A_n = 0. \quad \dots(10)$$

The equation (10) is known as the *auxiliary equation (A.E.)* and is obtained by putting $D = m$ and $D' = 1$ in $F(D, D') = 0$.

Let $m_1, m_2, \dots, m_n$ be n roots of A.E. (10). Two cases arise.

Case I. When $m_1, m_2, m_3, \dots, m_n$ are distinct. Then the part of C.F. corresponding to $m = m_r$ is $z = \phi_r(y + m_r x)$ for $r = 1, 2, 3, \dots, n$. Since (2) is linear, the sum of the solutions is also a solution.

$$\therefore \text{C.F. of (2)} = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x), \quad \dots(11)$$

where $\phi_1, \phi_2, \dots, \phi_n$ are arbitrary functions.

Case II. Repeated roots. Let m be repeated root of (10) and so consider

$$(D - mD')(D - mD')z = 0. \quad \dots(12)$$

$$\text{Let} \quad (D - mD')z = v. \quad \dots(13)$$

$$\text{Then,} \quad (12) \Rightarrow (D - mD')v = 0 \quad \text{or} \quad (\partial v / \partial x) - m(\partial v / \partial y) = 0, \quad \dots(14)$$

which is in Lagrange's form. Hence Lagrange's auxiliary equations for (14) are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dv}{0}. \quad \dots(15)$$

$$\text{As before, two independent integrals of (15) are} \quad y + mx = c_3 \quad \text{and} \quad v = c_4.$$

$$\therefore \quad v = \phi(y + mx) \quad \dots(16)$$

is a solution of (14), ϕ being an arbitrary function.

$$\text{Using (16), (13) becomes} \quad (\partial z / \partial x) - m(\partial z / \partial y) = \phi(y + mx) \quad \dots(17)$$

which is in Lagrange's form. Its Lagrange's auxiliary equations for (7) are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{\phi(y + mx)} \quad \dots(18)$$

$$\text{Taking the first two fractions of (18),} \quad dy + m dx = 0 \quad \text{so that} \quad y + mx = c_5. \quad \dots(19)$$

Taking the first and third fractions of (18) and using (19), we get

$$(dx)/1 = (dz)/\phi(c_5) \quad \text{so that} \quad dz - \phi(c_5) dx = 0.$$

$$\text{Integrating,} \quad z - x \phi(c_5) = c_6 \quad \text{or} \quad z - x \phi(y + mx) = c_6, \text{ using (19).} \quad \dots(20)$$

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From (19) and (20), the general solution of (12) is

$$z - x\phi(y + mx) = \psi(y + mx) \quad \text{or} \quad z = \psi(y + mx) + x\phi(y + mx), \quad \dots(21)$$

where ϕ and ψ are arbitrary functions. (21) is a part of C.F. corresponding to the two times repeated root m . In general, if a root ' m ' is repeated ' r ' times, the corresponding part of C.F. is

$$\phi_1(y + mx) + x\phi_2(y + mx) + x^2\phi_3(y + mx) + \dots + x^{r-1}\phi_r(y + mx).$$

4.4A Working rule for finding C.F. of linear homogeneous partial differential equation with constant coefficients

Step 1. Put the given equation in standard form $(A_0D^n + A_1D^{n-1}D' + \dots + A_nD'^n)z = f(x, y)$ (1)

Step 2. Replacing D by m and D' by 1 in the coefficients of z , we obtain auxiliary equation (A.E.) for (1) as $A_0m^n + A_1m^{n-1} + \dots + A_n = 0$ (2)

Step 3. The following six cases arise :

Case (i) Solve (2) for m . Let $m = m_1, m_2, \dots, m_n$ (different roots). Then

C.F. = $\phi_1(y + m_1x) + \phi_2(y + m_2x) + \dots + \phi_n(y + m_nx)$, where $\phi_1, \phi_2, \dots, \phi_n$ are arbitrary functions.

If in the above case (i), $m = a_1/b_1, a_2/b_2, \dots, a_n/b_n$, then

$$\text{C.F.} = \phi_1(b_1y + a_1x) + \phi_2(b_2y + a_2x) + \dots + \phi_n(b_ny + a_nx)$$

Further if $m = -(a_1/b_1), -(a_2/b_2), \dots, -(a_n/b_n)$, then

$$\text{C.F.} = \phi_1(b_1y - a_1x) + \phi_2(b_2y - a_2x) + \dots + \phi_n(b_ny - a_nx)$$

Case (ii) Solve (2) for m . Let $m = m'$ (repeated n times). Then corresponding to these roots

$$\text{C.F.} = \phi_1(y + m'x) + x\phi_2(y + m'x) + x^2\phi_3(y + m'x) + \dots + x^{n-1}\phi_n(y + m'x).$$

In the above case (ii), if $m = a/b$ (repeated n times), Then corresponding to these n roots,

$$\text{C.F.} = \phi_1(by + ax) + x\phi_2(by + ax) + x^2\phi_3(by + ax) + \dots + x^{n-1}\phi_n(by + ax)$$

And, if $m = -(a/b)$, (repeated n times), then

$$\text{C.F.} = \phi_1(by - ax) + x\phi_2(by - ax) + x^2\phi_3(by - ax) + \dots + x^{n-1}\phi_n(by - ax)$$

Case (iii) Corresponding to a non-repeated factor D on L.H.S. of (1),

the part of C.F. is taken as $\phi(y)$.

Case (iv) Corresponding to a repeated factor D^m on L.H.S. of (1), the part of C.F. is taken as

$$\phi_1(y) + x\phi_2(y) + x^2\phi_3(y) + \dots + x^{m-1}\phi_m(y).$$

Case (v) Corresponding to a non-repeated factor D' on L.H.S. of (1),

the part of C.F. is taken as $\phi(x)$.

Case (vi) Corresponding to a repeated factor D'^m on L.H.S. of (1), the part of C.F. is taken as

$$\phi_1(x) + y\phi_2(x) + y^2\phi_3(x) + \dots + y^{m-1}\phi_m(x).$$

4.4B Alternative working rule for finding C.F.

Let the given partial differential equation be $F(D, D')z = f(x, y)$. Factorize $F(D, D')$ into linear factors of the form $(bD - aD')$. Then we use the following results:

(i) Corresponding to each non-repeated factor $(bD - aD')$, the part of C.F. is taken as

$$\phi(by + ax).$$

(ii) Corresponding to a repeated factor $(bD - aD')^m$, the part of C.F. is taken as

$$\phi_1(by + ax) + x\phi_2(by + ax) + x^2\phi_3(by + ax) + \dots + x^{m-1}\phi_m(by + ax).$$

(iii) Corresponding to a non-repeated factor D , part of C.F. is taken as $\phi(y)$.

(iv) Corresponding to a repeated factor D^m , the part of C.F. is taken as

$$\phi_1(y) + x\phi_2(y) + x^2\phi_3(y) + \dots + x^{m-1}\phi_m(y).$$

(v) Corresponding to a non-repeated factor D' , part of C.F. is taken as $\phi(x)$.

(vi) Corresponding to a repeated factor D'^m , the part of C.F. is taken as

$$\phi_1(x) + y \phi_2(x) + y^2 \phi_3(x) + \dots + y^{m-1} \phi_m(x).$$

4.5 Solved examples based on Art. 4.4A and 4.4B

[Notations $p = \partial z / \partial x$, $q = \partial z / \partial y$, $r = \partial^2 z / \partial x^2$, $s = \partial^2 z / \partial x \partial y$ and $t = \partial^2 z / \partial y^2$ will be used]

Ex. 1. Solve (a) $r = a^2 t$.

[Avadh 2010; Rohilkhand 2010]

(b) $(\partial^2 z / \partial x^2) - (\partial^2 z / \partial y^2) = 0$.

[Kanpur 2016]

(c) $(D^2 - 3aDD' + 2a^2D'^2)z = 0$.

[Kanpur 2007; Meerut 2007]

Sol. (a) Given equation is $\partial^2 z / \partial x^2 = a^2 (\partial^2 z / \partial y^2)$ or $(D^2 - a^2 D'^2)z = 0$ (1)

The auxiliary equation of (1) is $m^2 - a^2 = 0$ so that $m = a, -a$.

$\therefore$ The general solution of (1) is $z = \text{C.F.} = \phi_1(y + ax) + \phi_2(y - ax)$,

where ϕ_1 and ϕ_2 are arbitrary functions.

(b) Proceed as in part (a).

Ans. $z = \phi_1(y + x) + \phi_2(y - x)$

(c) Proceed as in part (a).

Ans. $z = \phi_1(y + ax) + \phi_2(y + 2ax)$

Ex. 2. Solve (a) $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0$.

[Agra 2005]

(b) $(\partial^3 z / \partial x^3) - 7(\partial^3 z / \partial x \partial y^2) + 6(\partial^3 z / \partial y^3) = 0$.

[Bhopal 2010]

(c) $(D^3 - 3D^2D' + 2DD'^2)z = 0$.

[Meerut 2008, 12; Lucknow 2010]

Sol. (a) The auxiliary equation is $m^3 - 6m^2 + 11m - 6 = 0$

or $(m - 1)(m - 2)(m - 3) = 0$ so that $m = 1, 2, 3$.

$\therefore$ The general solution of the given equation is

$$z = \phi_1(y + x) + \phi_2(y + 2x) + \phi_3(y + 3x), \phi_1, \phi_2, \phi_3 \text{ being arbitrary functions.}$$

(b) The given equation can be written as $(D^3 - 7DD'^2 + 6D'^3)z = 0$ (1)

Its auxiliary equation is $m^3 - 7m + 6 = 0$ or $(m - 1)(m - 2)(m + 3) = 0$.

Hence $m = 1, 2, -3$ and so the general solution of (1) is

$$z = \phi_1(y + x) + \phi_2(y + 2x) + \phi_3(y - 3x), \phi_1, \phi_2, \phi_3 \text{ being arbitrary functions.}$$

(c) Proceed as above.

Ans. $z = \phi_1(y) + \phi_2(y + x) + \phi_3(y + 2x)$.

Ex. 3. (a) Solve $2r + 5s + 2t = 0$.

[Meerut 2011]

(b) $2(\partial^2 z / \partial x^2) - 3(\partial^2 z / \partial x \partial y) - 2(\partial^2 z / \partial y^2) = 0$.

Sol. (a) Now, $r = \partial^2 z / \partial x^2 = D^2 z$, $s = \partial^2 z / \partial x \partial y = DD' z$ and $t = \partial^2 z / \partial y^2 = D'^2 z$. Hence the given equation can be re-written as $(2D^2 + 5DD' + 2D'^2)y = 0$ (1)

Its auxiliary equation is $2m^2 + 5m + 2 = 0$ or $(2m + 1)(m + 2) = 0$.

So $m = -1/2, -2$ and hence the general solution of (1) is

$$z = \phi_1(2y - x) + \phi_2(y - 2x), \phi_1 \text{ and } \phi_2 \text{ being arbitrary functions.}$$

Alternative method : (1) can be re-written as $(2D + D')(D + 2D') = 0$.

So by using the alternative working rule 4.4B, the general solution of (1) is

$$z = \phi_1(2y - x) + \phi_2(y - 2x), \phi_1, \phi_2 \text{ being arbitrary functions.}$$

(b) Proceed as in part (a).

Ans. $z = \phi_1(2y - x) + \phi_2(y + 2x)$

Ex. 4. Solve (a) $r + t + 2s = 0$

[Kanpur 2009]

(b) $25r - 40s + 16t = 0$

[Jabalpur 2002; Sagar 2004];

(c) $(4D^2 + 12DD' + 9D'^2)z = 0$.

[Kanpur 2008; Indore 2004]

Sol. (a) Here $r = \partial^2 z / \partial x^2 = D^2 z$, $t = \partial^2 z / \partial y^2 = D'^2 z$, $s = \partial^2 z / \partial x \partial y = DD' z$.

So the given equation becomes $(D^2 + D'^2 + 2DD')z = 0$ (1)

Its auxiliary equation is $m^2 + 1 + 2m = 0$ or $(m + 1)^2 = 0$ or $m = -1, -1$.

So the general solution of (1) is $z = \phi_1(y - x) + x\phi_2(y - x)$, ϕ_1 and ϕ_2 being arbitrary functions.

(b) Here $r = \partial^2 z / \partial x^2 = D^2 z$, $s = \partial^2 z / \partial x \partial y = DD' z$, $t = \partial^2 z / \partial y^2 = D'^2 z$. So the given equation becomes $(25D^2 - 40DD' + 16D'^2)z = 0$ (1)

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Its auxiliary equation is $25m^2 - 40m + 16 = 0$ or $(5m - 4)^2 = 0$ so that $m = 4/5, 4/5$.
Hence the general solution of (1) is $z = \phi_1(5y + 4x) + x\phi_2(5y + 4x)$.

Alternative method (1) may be written as $(5D - 4D')^2 z = 0$.

Using result (ii) of Art 4.4B, the solution of (1) is

$$z = \phi_1(5y + 4x) + x\phi_2(5y + 4x), \phi_1, \phi_2 \text{ being arbitrary functions.}$$

(c) Proceed as in part (b).

$$\text{Ans. } z = \phi_1(2y - 3x) + x\phi_2(2y - 3x)$$

Ex. 5. Solve (a) $(D^3 - 4D^2D' + 4DD'^2)z = 0$.

[Bhopal 2000, 03]

(b) $(D^4 - 2D^3D' + 2DD'^3 - D'^4)z = 0$.

[Lucknow 2009; Bilaspur 2004]

(c) $(D^4 + D'^2 - 2D^2D'^2)z = 0$.

(d) $(D^3 - 3D^2D' + 3DD'^2 - D'^3)z = 0$.

Sol. (a) The auxiliary equation of the given equation is

$$m^3 - 4m^2 + 4m = 0 \quad \text{or} \quad m(m - 2)^2 = 0 \quad \text{so that} \quad m = 0, 2, 2$$

Hence the general solution of the given equation is

$$z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x), \phi_1, \phi_2, \phi_3 \text{ being arbitrary functions.}$$

(b) The auxiliary equation of the given equation is

$$m^4 - 2m^3 + 2m - 1 = 0 \quad \text{or} \quad (m + 1)(m - 1)^3 = 0 \quad \text{so that} \quad m = -1, 1, 1, 1.$$

Hence the general solution of the given equation is

$z = \phi_1(y - x) + \phi_2(y + x) + x\phi_3(y + x) + x^2\phi_4(y + x)$, where ϕ_1, ϕ_2, ϕ_3 and ϕ_4 are arbitrary functions.

(c) Try yourself.

$$\text{Ans. } z = \phi_1(y + x) + x\phi_2(y + x) + \phi_3(y - x) + x\phi_4(y - x)$$

(d) Proceed as in part (a).

$$\text{Ans. } z = \phi_1(y + x) + x\phi_2(y + x) + x^2\phi_3(y + x)$$

Ex. 6. Solve (a) $(D^3D'^2 + D^2D'^3)z = 0$.

(b) $(D^3D' - 4D^2D'^2 + 4DD'^3)z = 0$.

Sol. (a) The given equation can be re-written as

$$D^2D'^2(D + D')z = 0. \quad \dots(1)$$

Hence using the alternative method 4.4B for C.F., the general solution is

$$z = \phi_1(y) + x\phi_2(y) + \phi_3(x) + y\phi_4(x) + \phi_5(y - x), \text{ where } \phi_1, \phi_2, \phi_3, \phi_4 \text{ and } \phi_5 \text{ are arbitrary functions.}$$

(b) The given equation can be re-written as

$$DD'(D^2 - 4DD' + 4D'^2) = 0 \quad \text{Or} \quad DD'(D - 2D')^2 = 0.$$

Using the working rule 4.4B, the required general solution is

$$z = \phi_1(y) + \phi_2(x) + \phi_3(y + 2x) + x\phi_4(y + 2x), \text{ where } \phi_1, \phi_2, \phi_3 \text{ and } \phi_4 \text{ are arbitrary functions.}$$

Ex. 7. Solve (a) $(\partial^4 z / \partial x^4) - (\partial^4 z / \partial y^4) = 0$

(b) $(D^4 + D'^4)z = 0$.

Sol. (a) Rewriting, the given equation is

$$(D^4 - D'^4)z = 0. \quad \dots(1)$$

Its auxiliary equation is $m^4 - 1 = 0$ or $(m^2 - 1)(m^2 + 1) = 0 \Rightarrow m = 1, -1, i, -i$.

Hence the general solution of (1) is $z = \phi_1(y - x) + \phi_2(y + x) + \phi_3(y + ix) + \phi_4(y - ix)$,

where ϕ_1, ϕ_2, ϕ_3 and ϕ_4 are arbitrary functions.

(b) The auxiliary equation of the given equation is $m^4 + 1 = 0$ or $(m^2 + 1)^2 - 2m^2 = 0$

$$\text{or} \quad (m^2 + 1)^2 - (m\sqrt{2})^2 = 0 \quad \text{or} \quad (m^2 + 1 + \sqrt{2}m)(m^2 + 1 - \sqrt{2}m) = 0$$

$$\text{so that} \quad m^2 + \sqrt{2}m + 1 = 0 \quad \text{or} \quad m^2 - \sqrt{2}m + 1 = 0 \Rightarrow m = (-1 \pm i)/\sqrt{2}, (1 \pm i)/\sqrt{2}.$$

$$\text{Let } z_1 = (-1 + i)/\sqrt{2} \text{ and } z_2 = (1 + i)/\sqrt{2}, \text{ then, } m = z_1, \bar{z}_1, z_2, \bar{z}_2,$$

where $\bar{z}_1$ and $\bar{z}_2$ denote complex conjugates of z_1 and z_2 respectively.

Hence the general solution of the given equation is

$$z = \phi_1(y + z_1 x) + \phi_2(y + \bar{z}_1 x) + \phi_3(y + z_2 x) + \phi_4(y + \bar{z}_2 x), \text{ where } \phi_1, \phi_2, \phi_3, \phi_4 \text{ are arbitrary functions.}$$

4.6 Particular integral (P.I.) of homogeneous linear partial differential equation given by

$$F(D, D')y = f(x, y) \quad \dots (1)$$

The inverse operator $1/F(D, D')$ of the operator $F(D, D')$ is defined by the following identity

$$F(D, D') \left(\frac{1}{F(D, D')} f(x, y) \right) = f(x, y)$$

$$\therefore \text{Particular integral (P.I.) of (1)} = \frac{1}{F(D, D')} f(x, y)$$

In what follows we shall treat the symbolic functions of D and D' as we do for the symbolic functions of D alone in ordinary differential equations. Thus it will be factorized and resolved into partial fractions or expanded in an infinite series as the case may be. The reader is advised to note carefully the following results :

(i) $D, D^2, \dots$ will stand for differentiating partially with respect to x once, twice and so on.

For example, $Dx^4y^5 = \frac{\partial}{\partial x} x^4 y^5 = 4x^3 y^5$; $D^2x^4y^5 = \frac{\partial^2}{\partial x^2} x^4 y^5 = 12x^2 y^5$.

(ii) $D', D'^2, \dots$ will stand for differentiating partially with respect to y once, twice and so on.

For example, $D'x^4y^5 = \frac{\partial}{\partial y} x^4 y^5 = 5x^4 y^4$; $D'^2x^4y^5 = \frac{\partial^2}{\partial y^2} x^4 y^5 = 20x^4 y^3$.

(iii) $1/D, 1/D^2, \dots$ will stand for integrating partially with respect to x once, twice and so on.

For example, $\frac{1}{D} x^4 y^5 = \int x^4 y^5 dx = \frac{x^5 y^5}{5}$; $\frac{1}{D^2} x^4 y^5 = \iint x^4 y^5 dx dx = \frac{x^6 y^5}{30}$

(iv) $1/D', 1/D'^2, \dots$ will stand for integrating partially with respect to y once, twice and so on.

For example, $\frac{1}{D'} x^4 y^5 = \int x^4 y^5 dy = \frac{x^4 y^6}{6}$; $\frac{1}{D'^2} x^4 y^5 = \iint x^4 y^5 dy dy = \frac{x^4 y^7}{42}$.

4.7 Short methods of finding the P.I. in certain cases

Before taking up the general method for finding P.I. of $F(D, D')z = f(x, y)$ we begin with cases when $f(x, y)$ is in two special forms. The methods corresponding to these forms are much shorter than the general methods to be discussed in Art. 4.12.

4.8 Short Method I. When $f(x, y)$ is of the form $f(ax + by)$

The method under consideration is based on the following theorem.

Theorem I. If $F(D, D')$ be homogeneous function of D and D' of degree n , then

$$\frac{1}{F(D, D')} \phi^{(n)}(ax + by) = \frac{1}{F(a, b)} \phi(ax + by),$$

provided $F(a, b) \neq 0$, $\phi^{(n)}$ being the n th derivative of ϕ w.r.t. $ax + by$ as a whole.

Proof. By direct differentiation, we have $D'\phi(ax + by) = \alpha'\phi^{(r)}(ax + by)$,
 $D'^s\phi(ax + by) = b^s\phi^{(s)}(ax + by)$ and $D'D'^s\phi(ax + by) = \alpha'b^s\phi^{(r+s)}(ax + by)$.

Since $F(D, D')$ is homogeneous function of degree n , so we have

$$F(D, D')\phi(ax + by) = F(a, b)\phi^{(n)}(ax + by). \quad \dots(1)$$

Operating both sides of (1) by $1/F(D, D')$, we have

$$\phi(ax + by) = F(a, b) \frac{1}{F(D, D')} \phi^{(n)}(ax + by). \quad \dots(2)$$

Since $F(a, b) \neq 0$, dividing both sides of (2) by $F(a, b)$, we get

$$\frac{1}{F(D, D')} \phi^{(n)}(ax + by) = \frac{1}{F(a, b)} \phi(ax + by). \quad \dots(3)$$

An important deduction from result (3) : Putting $ax + by = v$, (3) gives

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$$\frac{1}{F(D, D')} \phi^{(n)}(v) = \frac{1}{F(a, b)} \phi(v). \quad \dots(4)$$

Integrating both sides of (4) n times w.r.t. ' v ', we have

$$\frac{1}{F(D, D')} \phi(v) = \frac{1}{F(a, b)} \int \int \dots \int \phi(v) dv dv \dots dv, \quad \text{where} \quad v = ax + by.$$

Exceptional case when $F(a, b) = 0$. When $F(a, b) = 0$, then the above theorem does not hold good. In such a case the new method is based on the following theorem. Note that $F(a, b) = 0$ if and only if $(bD - aD')$ is a factor $F(D, D')$.

Theorem II.
$$\frac{1}{(bD - aD')^n} \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by).$$

Proof. Consider the equation $(bD - aD')z = x^r \phi(ax + by) \quad \dots(1)$

or $bp - aq = x^r \phi(ax + by). \quad \dots(2)$

Lagrange's subsidiary equations for (2) are
$$\frac{dx}{b} = \frac{dy}{-a} = \frac{dz}{x^r \phi(ax + by)}. \quad \dots(3)$$

Taking the first two fractions of (3), $adx + bdy = 0$ so that $ax + by = c_1 \quad \dots(4)$

Taking the first and third members of (4) and using (4), we get

$$\frac{dx}{b} = \frac{dz}{x^r \phi(c_1)} \quad \text{or} \quad dz = \frac{x^r \phi(c_1)}{b} dx.$$

Integrating,
$$z = \frac{x^{r+1} \phi(c_1)}{b(r+1)} = \frac{x^{r+1} \phi(ax + by)}{b(r+1)}, \text{ by (4).} \quad \dots(5)$$

(5) is a solution of (1).

Now, from (1),
$$z = \frac{1}{(bD - aD')^n} x^r \phi(ax + by). \quad \dots(6)$$

From (5) and (6),
$$\frac{1}{(bD - aD')^n} x^r \phi(ax + by) = \frac{x^{r+1}}{b(r+1)} \phi(ax + by). \quad \dots(7)$$

Hence, if
$$z = \frac{1}{(bD - aD')^n} \phi(ax + by), \quad \text{then we have}$$

$$\begin{aligned} z &= \frac{1}{(bD - aD')^{n-1}} \left[\frac{1}{(bD - aD')} x^0 \phi(ax + by) \right], \text{ as } x^0 = 1 \\ &= \frac{1}{(bD - aD')^{n-1}} \frac{x}{b} \phi(ax + by), \text{ using (7) for } r = 0 \\ &= \frac{1}{b} \frac{1}{(bD - aD')^{n-2}} \left[\frac{1}{(bD - aD')} x \phi(ax + by) \right] \\ &= \frac{1}{b} \frac{1}{(bD - aD')^{n-2}} \frac{x^2}{2b} \phi(ax + by), \text{ using (7) for } r = 1 \\ &= \frac{1}{2! b^2} \frac{1}{(bD - aD')^{n-2}} x^2 \phi(ax + by) \\ &= \frac{1}{n! b^n} \frac{1}{(bD - aD')^{n-n}} x^n \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by) \end{aligned}$$

[after repeated use of (7) for $n - 2$ times more]

Working rule for finding particular integral where $f(x, y) = \phi(ax + by)$.

The following rules will be used depending upon the situation in hand.

Formula (i). When $F(a, b) \neq 0$ and $F(D, D')$ is a homogeneous function of degree n , then

$$\text{P.I.} = \frac{1}{F(D, D')} \phi(ax + by) = \frac{1}{F(a, b)} \int \dots \int f(v) dv dv \dots dv, \quad \text{where } v = ax + by$$

Note that R.H.S. contains a multiple integral of n th order.

Formula (ii). When $F(a, b) = 0$, we have

$$\frac{1}{(bD - aD')^n} \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by).$$

4.9 Solved Examples based on short method I of Art. 4.8

Ex. 1. Solve $(D^2 + 3DD' + 2D'^2)z = x + y$. [Agra 2011, Meerut 2010; Rohilkhand 2010; Kanpur 2014]

Sol. The auxiliary equation of the given equation is $m^2 + 3m + 2 = 0$ giving $m = -1, -2$.

$\therefore$ C.F. = $\phi_1(y - x) + \phi_2(y - 2x)$, ϕ_1, ϕ_2 being arbitrary functions.

Now,
$$\text{P.I.} = \frac{1}{D^2 + 3DD' + 2D'^2} (x + y)$$

$$= \frac{1}{1^2 + 3 \times 1 \times 1 + 2 \times 1^2} \int \int v dv dv, \text{ where } v = x + y, \text{ using formula (i) of working rule}$$

$$= \int (v^2 / 2) dv = (1/6) \times (v^3 / 6) = (1/36) \times (x + y)^3.$$

Hence the required general solution is $z = \text{C.F.} + \text{P.I.}, i.e.,$

$$z = \phi_1(y - x) + \phi_2(y - 2x) + (1/36) \times (x + y)^3.$$

Ex. 2. Solve (a) $(2D^2 - 5DD' + 2D'^2)z = 24(y - x)$. [Vikram 2000]

(b) $(\partial^2 V / \partial x^2) + (\partial^2 V / \partial y^2) = 12(x + y)$ [Nagpur 2010]

(c) $(D^2 + D'^2)z = 30(2x + y)$. [Agra 2012; Sagar 2004]

(d) $(\partial^2 z / \partial x^2) + 2(\partial^2 z / \partial x \partial y) + (\partial^2 z / \partial y^2) = 2x + 3y$.

(e) $(D^2 + 3DD' + 2D'^2)z = 2x + 3y$. [Kurukshetra 2005]

(f) $r + s - 2t = (2x + y)^{1/2}$. [Lucknow 2010]

Sol. (a) The auxiliary of the given equation is $2m^2 - 5m + 2 = 0$ giving $m = 1/2, 2$.

$\therefore$ C.F. = $\phi_1(2y + x) + \phi_2(y + 2x)$, ϕ_1, ϕ_2 being arbitrary functions.

Now,
$$\text{P.I.} = \frac{1}{2D^2 - 5DD' + 2D'^2} 24(y - x) = 24 \frac{1}{2D^2 - 5DD' + 2D'^2} (y - x)$$

$$= \frac{24}{2 \times (-1)^2 - 5 \times (-1) \times 2 + 2 \times 2^2} \int \int v dv dv, \text{ where } v = y - x, \text{ using formula (i) of working rule}$$

$$= (24/20) \times \int (v^2 / 2) dv = (6/5) \times (v^3 / 6) = (1/5) \times (y - x)^3.$$

Hence the required general solution is $z = \phi_1(2y + x) + \phi_2(y + 2x) + (y - x)^3 / 5$.

(b) The given equation can be written as $(D^2 + D'^2)V = 12(x + y)$ (1)

Its auxiliary equation is $m^2 + 1 = 0$ so that $m = \pm i$.

$\therefore$ C.F. = $\phi_1(y + ix) + \phi_2(y - ix)$, ϕ_1, ϕ_2 being arbitrary functions.

Now,
$$\text{P.I.} = \frac{1}{D^2 + D'^2} 12(x + y) = 12 \frac{1}{D^2 + D'^2} (x + y) = \frac{12}{1^2 + 1^2} \int \int v dv dv$$

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$$= 6 \int (v^2/2) dv = v^3 = (x+y)^3.$$

Hence the required general solution is

$$V = \phi_1(y+ix) + \phi_2(y-ix) + (x+y)^3.$$

(c) Try yourself.

$$\text{Ans. } z = \phi_1(y+ix) + \phi_2(y-ix) + (2x+y)^3$$

(d) Try yourself.

$$\text{Ans. } z = \phi_1(y-x) + x\phi_2(y-x) + (1/150) \times (2x+3y)^3$$

(e) Proceed as in part (d).

$$\text{Ans. } y = \phi_1(y-x) + \phi_2(y-2x) + (1/240) \times (2x+3y)^3$$

(f) Since $r = \partial^2 z / \partial x^2$, $s = \partial^2 z / \partial x \partial y$, $t = \partial^2 z / \partial y^2$, the given equation can be re-written as

$$\partial^2 z / \partial x^2 + \partial^2 z / \partial x \partial y - 2(\partial^2 z / \partial y^2) = (2x+y)^{1/2} \quad \text{or} \quad (D^2 + DD' - 2D'^2)z = (2x+y)^{1/2}.$$

Its auxiliary equation is

$$m^2 + m - 2 = 0$$

so that

$$m = 1, -2.$$

$\therefore$ C.F. = $\phi_1(y+x) + \phi_2(y-2x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{P.I.} = \frac{1}{D^2 + DD' - 2D'^2} (2x+y)^{1/2} = \frac{1}{2^2 + 2 \times 1 - 2 \times 1^2} \iint v^{1/2} dv dv, \text{ where } v = 2x+y$$

$$= \frac{1}{4} \int \frac{2}{3} v^{3/2} dv = \frac{1}{4} \times \frac{2}{3} \times \frac{2}{5} v^{5/2} = \frac{1}{15} (2x+y)^{5/2}$$

Hence the required general solution is

$$z = \phi_1(y+x) + \phi_2(y-2x) + (1/15)(2x+y)^{5/2}.$$

Ex. 3. Solve (a) $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$. [Bhopal 2010; Meerut 2012; Purvanchal 2007;

K.V. Kurkshetra 2005]

(b) $(D^2 - 2DD' + D'^2)z = e^{x+2y}$.

[Kanpur 2005]

Sol. (a) Here auxiliary equation is

$$m^2 + 2m + 1 = 0$$

so that

$$m = -1, -1.$$

$\therefore$ C.F. = $\phi_1(y-x) + x\phi_2(y-x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{P.I.} = \frac{1}{D^2 + 2DD' + D'^2} e^{2x+3y} = \frac{1}{(D+D')^2} e^{2x+3y} = \frac{1}{(2+3)^2} \iint e^v dv dv, \text{ where } v = 2x+3y$$

$$= (1/25) \times \int e^v dv = (1/25) \times e^v = (1/25) \times e^{2x+3y}$$

$\therefore$ Solution is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y-x) + x\phi_2(y-x) + (1/25) \times e^{2x+3y}.$$

(b) Proceed as in part (a).

$$\text{Ans. } z = \phi_1(y+x) + x\phi_2(y+x) + e^{x+2y}$$

Ex. 4. Solve (a) $r - 2s + t = \sin(2x+3y)$.

[Meerut 2013; Indore 2002; Kanpur 2010]

(b) $(D^3 - 4D^2D' + 4DD'^2)z = \cos(2x+3y)$.

(c) $(D^3 - 3DD'^2 - 2D'^3)z = \cos(x+2y)$.

Sol. (a) Since $r = \partial^2 z / \partial x^2$, $s = \partial^2 z / \partial x \partial y$, $t = \partial^2 z / \partial y^2$, the given equation becomes

$$\partial^2 z / \partial x^2 - 2(\partial^2 z / \partial x \partial y) + \partial^2 z / \partial y^2 = \sin(2x+3y) \quad \text{or} \quad (D^2 - 2DD' + D'^2)z = \sin(2x+3y).$$

Its auxiliary equation is

$$m^2 - 2m + 1 = 0$$

so that

$$m = 1, 1.$$

$\therefore$ C.F. = $\phi_1(y+x) + x\phi_2(y+x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{P.I.} = \frac{1}{(D-D')^2} \sin(2x+3y) = \frac{1}{(2-3)^2} \iint \sin v dv dv, \text{ where } v = 2x+3y$$

$$= - \int \cos v dv = -\sin v = -\sin(2x+3y).$$

Hence the required general solution is

$$z = \phi_1(y+x) + x\phi_2(y+x) - \sin(2x+3y).$$

(b) Proceed as in part (a). **Ans.** $z = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x) - (1/32) \times \sin(2x+3y)$

(c) Proceed as in part (a). **Ans.** $z = \phi_1(y-x) + x\phi_2(y-x) + \phi_3(y+2x) + (1/27) \times \sin(x+2y)$

Ex. 5(a). Solve $\partial^2 z / \partial x^2 + \partial^2 z / \partial y^2 = \cos mx \cos ny$. [Kanpur 2015; Nagpur 2010; Lucknow 2013]

Sol. Given equation can be written as

$$(D^2 + D'^2)z = \cos mx \cos ny.$$

Its auxiliary equation is

$$m^2 + 1 = 0$$

so that

$$m = \pm i.$$

∴ C.F. = $\phi_1(y + ix) + \phi_2(y - ix)$, ϕ_1 and ϕ_2 being arbitrary functions.

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + D'^2} \cos mx \cos ny = \frac{1}{D^2 + D'^2} \frac{\cos(mx + ny) + \cos(mx - ny)}{2} \\ &= \frac{1}{2} \frac{1}{D^2 + D'^2} \cos(mx + ny) + \frac{1}{2} \frac{1}{D^2 + D'^2} \cos(mx - ny) \\ &= \frac{1}{2} \frac{1}{m^2 + n^2} \iint \cos v \, dv \, dv + \frac{1}{2} \frac{1}{m^2 + (-n)^2} \iint \cos u \, du \, du, \\ &\quad \text{where } v = mx + ny \text{ and } u = mx - ny \\ &= \frac{1}{2} \frac{1}{m^2 + n^2} \int \sin v \, dv + \frac{1}{2} \frac{1}{m^2 + n^2} \int \sin u \, du = \frac{1}{2} \frac{1}{m^2 + n^2} [-\cos v - \cos u] \\ &= -\frac{1}{2(m^2 + n^2)} [\cos(mx + ny) + \cos(mx - ny)], \text{ as } v = mx + ny, u = mx - ny \\ &= -\frac{1}{2(m^2 + n^2)} \times 2 \cos mx \cos ny = -(m^2 + n^2)^{-1} \cos mx \cos ny. \end{aligned}$$

Hence the required general solution is $z = \phi_1(y + ix) + \phi_2(y - ix) - (m^2 + n^2)^{-1} \cos mx \cos ny$.

Ex. 5. (b) Solve $\partial^2 z / \partial x^2 + \partial^2 z / \partial y^2 = \cos mx \sin ny$. [Ravishankar 2001]

Sol. Do like Ex. 5(a) **Ans.** $z = \phi_1(y + ix) + \phi_2(y - ix) + (\sin mx + \sin ny) / (m^2 + n^2)$

Ex. 6. Solve the following partial differential equations:

(a) $(D^2 - 2DD' + D'^2)z = \tan(y + x)$ or $(D - D')^2 z = \tan(y + x)$

(b) $(D^2 - 2aDD' + a^2D'^2)z = f(y + ax)$ or $(D - aD')^2 z = f(y + ax)$.

(c) $4r - 4s + t = 16 \log(x + 2y)$. [Agra 2009, 2012; Meerut 2009; Lucknow 2007]

Sol. (a) Here auxiliary equation is $(m - 1)^2 = 0$ so that $m = 1, 1$.

∴ C.F. = $\phi_1(y + x) + x\phi_2(y + x)$, where ϕ_1 and ϕ_2 are arbitrary functions.

$$\text{Now, P.I.} = \frac{1}{(D - D')^2} \tan(y + x) = \frac{x^2}{1^2 \times 2!} \tan(y + x) = \frac{x^2}{2} \tan(y + x)$$

[Using formula (ii) of working rule with $a = 1, b = 1, n = 2$]

Hence the required general solution is $z = \phi_1(y + x) + x\phi_2(y + x) + (x^2/2) \times \tan(y + x)$.

(b) Here auxiliary equation is $(m - a)^2 = 0$ so that $m = a, a$.

∴ C.F. = $\phi_1(y + ax) + x\phi_2(y + ax)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{P.I.} = \frac{1}{(D - aD')^2} f(y + ax) + \frac{x^2}{1^2 \times 2!} f(y + ax) = \frac{x^2}{2} f(y + ax).$$

[using formula (ii) of working rule with $a = a, b = 1, n = 2$]

∴ General solution is $z = \phi_1(y + ax) + x\phi_2(y + ax) + (x^2/2) \times f(y + ax)$.

(c) Since $r = \partial^2 z / \partial x^2, s = \partial^2 z / \partial x \partial y, t = \partial^2 z / \partial y^2$, the given equation becomes

$$4(\partial^2 z / \partial x^2) - 4(\partial^2 z / \partial x \partial y) + (\partial^2 z / \partial y^2) = 16 \log(x + 2y) \quad \text{or} \quad (4D^2 - 4DD' + D'^2)z = 16 \log(x + 2y)$$

Its auxiliary equation is $4m^2 - 4m + 1 = 0$ so that $m = 1/2, 1/2$.

∴ C.F. = $\phi_1(2y + x) + x\phi_2(2y + x)$, ϕ_1 and ϕ_2 being arbitrary functions.

$$\text{Now, P.I.} = \frac{1}{(2D - D')^2} 16 \log(x + 2y) = 16 \times \frac{x^2}{2^2 \times 2!} \log(x + 2y) = 2x^2 \log(x + 2y)$$

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[using formula (ii) of working rule with $a = 1, b = 2, n = 2$]

∴ The required solution is $z = \phi_1(2y + x) + x\phi_2(2y + x) + 2x^2 \log(x + 2y)$.

Ex. 7. Solve the following partial differential equations :

(a) $(D^3 - 2D^2D' - DD'^2 + 2D'^3)z = e^{x+y}$. [Bhopal 2000, 03, Meerut 2007; Jabalpur 2004]

(b) $r - t = x - y$. or $\partial^2 z / \partial x^2 - \partial^2 z / \partial y^2 = x - y$ [Agra 2012]

(c) $2r - s - 3t = 5e^x/e^y$. [Rohilkhand 2009; Indore 2003; Jiwaji 2003]

(d) $r + 5s + 6t = (y - 2x)^{-1}$ or $(\partial^2 z / \partial x^2) + 5(\partial^2 z / \partial x \partial y) + 6(\partial^2 z / \partial y^2) = 1 / (y - 2x)$

[Meerut 2012; Agra 2009; Indore 2000; Garhwal 2005]

Sol. (a) Here auxiliary equation is $m^3 - 2m^2 - m + 2 = 0$ or $m^2(m - 2) - (m - 2) = 0$
or $(m^2 - 1)(m - 2) = 0$ so that $m = 2, 1, -1$.

∴ C.F. = $\phi_1(y + 2x) + \phi_2(y + x) + \phi_3(y - x)$, ϕ_1, ϕ_2, ϕ_3 being arbitrary functions.

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^3 - 2D^2D' - DD'^2 + 2D'^3} e^{x+y} = \frac{1}{(D - D')} \left\{ \frac{1}{(D^2 - DD' - 2D'^2)} e^{x+y} \right\} \\ &= \frac{1}{D - D'} \cdot \frac{1}{1^2 - (1 \times 1) - (2 \times 1)^2} \int e^v dv, \text{ where } v = x + y, \text{ using formula (i) of working rule} \\ &= \frac{1}{D - D'} \left(-\frac{1}{2} \right) e^v = -\frac{1}{2} \frac{1}{D - D'} e^{x+y} = -\frac{1}{2} \times \frac{x}{1^1 \times 1!} e^{x+y} = -\frac{x}{2} e^{x+y} \end{aligned}$$

[Using formula (ii) with $a = 1, b = 1, n = 1$]

∴ Required solution is $z = \phi_1(y + 2x) + \phi_2(y + x) + \phi_3(y - x) - (x/2) \times e^{x+y}$.

(b) The given equation can be re-written as

$$(\partial^2 z / \partial x^2) - (\partial^2 z / \partial y^2) = x - y \quad \text{or} \quad (D^2 - D'^2)z = x - y. \quad \dots(1)$$

Its auxiliary equation is $m^2 - 1 = 0$ so that $m = 1, -1$.

∴ C.F. = $\phi_1(y + x) + \phi_2(y - x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\begin{aligned} \text{Now, P.I.} &= \frac{1}{D^2 - D'^2} (x - y) = \frac{1}{D + D'} \cdot \frac{1}{D - D'} (x - y) = \frac{1}{D + D'} \cdot \frac{1}{1 - (-1)} \int v dv, \text{ where } v = x - y. \\ &= \frac{1}{2} \frac{1}{D + D'} \frac{v^2}{2} = \frac{1}{4} \frac{1}{D + D'} (x - y)^2 = -\frac{1}{4} \frac{1}{[(-1) \times D - 1 \times D']^1} (x - y)^2 \\ &= -\frac{1}{4} \frac{x}{(-1)^1 \times 1!} (x - y)^2 = \frac{x}{4} (x - y)^2, \text{ by formula (ii) with } a = 1, b = -1, n = 1. \end{aligned}$$

∴ The required solution is $z = \phi_1(y + x) + \phi_2(y - x) + (x/4) \times (x - y)^2$.

(c) Try yourself. **Ans.** $z = \phi_1(y - x) + \phi_2(2y + 3x) + x e^{x/y}$

(d) Try yourself. **Ans.** $z = \phi_1(y - 2x) + \phi_2(y - 3x) + x \log(y - 2x)$.

Ex. 8. Solve the following partial differential equation:

(a) $(D^3 - 4D^2D' + 4DD'^2)z = 4 \sin(2x + y)$.

[Rohilkhand 2011; Indore 2004; Rewa 2000, 01; MDU Rohtak 2004]

(b) $(D^3 - 4D^2D' + 4DD'^2)z = \cos(2x + y)$.

(c) $(D^3 - 4D^2D' + 4DD'^2)z = \sin(y + 2x)$.

Sol. (a) Here the auxiliary equation is $m^3 - 4m^2 + 4m = 0$ or $m(m^2 - 4m + 4) = 0$
or $m(m - 2)^2 = 0$ so that $m = 0, 2, 2$.

∴ C.F. = $\phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x)$, ϕ_1, ϕ_2, ϕ_3 being arbitrary functions.

$$\begin{aligned}
 \text{Now, P.I.} &= \frac{1}{D^3 - 4D^2D' + 4DD'^2} 4 \sin(2x + y) = 4 \frac{1}{(D - 2D')^2} \left\{ \frac{1}{D} \sin(2x + y) \right\} \\
 &= 4 \frac{1}{(D - 2D')^2} \left\{ -\frac{1}{2} \cos(2x + y) \right\}, \quad \begin{array}{l} \text{since } 1/D \text{ stands for integration} \\ \text{w.r.t. } x \text{ treating } y \text{ as constant.} \end{array} \\
 &= -2 \frac{1}{(D - 2D')^2} \cos(2x + y) = -2 \frac{x^2}{1^2 \times 2!} \cos(2x + y),
 \end{aligned}$$

[Using formula (ii) with $a = 2, b = 1, n = 2$]So the required solution is $y = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x) - x^2 \cos(2x + y)$.(b) Try yourself **Ans.** $z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x) + (x^2/4) \times \sin(2x + y)$ (c) Try yourself **Ans.** $z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x) - (x^2/4) \times \cos(2x + y)$ **Ex. 9.** Solve the following partial differential equations :(a) $(D^2 - 3DD' + 2D'^2)z = e^{2x-y} + e^{x+y} + \cos(x + 2y)$.(b) $(D^2 - 3DD' + 2D'^2)z = e^{2x-y} + \cos(x + 2y)$ [Delhi B.Sc. (Hons.) 2006](c) $(D^3 - 4D^2D' + 5DD'^2 - 2D'^3)z = e^{y+2x} + (y+x)^{1/2}$.**Sol.** (a) Here auxiliary equation is $m^2 - 3m + 2 = 0$ so that $m = 1, 2$. $\therefore$ C.F. = $\phi_1(y + x) + \phi_2(y + 2x)$, ϕ_1, ϕ_2 being arbitrary functions. ... (1)Now, P.I. corresponds to e^{2x-y}

$$\begin{aligned}
 &= \frac{1}{D^2 - 3DD' + 2D'^2} e^{2x-y} = \frac{1}{2^2 - 3 \times 2 \times (-1) + 2 \times (-1)^2} \iint e^v dv dv, \text{ where } v = 2x - y \\
 &= (1/12) \times \int e^v dv = (1/12) \times e^v = (1/12) \times e^{2x-y}.
 \end{aligned}$$

... (2)

$$\text{P.I. corresponding to } e^{x+y} = \frac{1}{D^2 - 3DD' + 2D'^2} e^{x+y} = \frac{1}{D - D'} \left\{ \frac{1}{D - 2D'} e^{x+y} \right\}$$

$$= \frac{1}{D - D'} \left\{ \frac{1}{1 - (2 \times 1)} \int e^v dv \right\}, \text{ where } v = x + y, \text{ using formula (i)}$$

$$= -\frac{1}{D - D'} e^v = -\frac{1}{(D - D')^1} e^{x+y} = -\frac{x}{1^1 \times 1!} e^{x+y} = -x e^{x+y}.$$

... (3)

[Using formula (ii) with $a = b = 1, n = 1$]Finally, P.I. corresponding to $\cos(x + 2y)$

$$\begin{aligned}
 &= \frac{1}{D^2 - 3DD' + 2D'^2} \cos(x + 2y) = \frac{1}{1^2 - (3 \times 1 \times 2) + (2 \times 2^2)} \iint \cos v dv dv, \text{ where } v = x + 2y \\
 &= (1/3) \times \int \sin v dv = -(1/3) \times \cos v = -(1/3) \times \cos(x + 2y)
 \end{aligned}$$

... (4)

From (1), (2), (3) and (4), the required solution is $z = \text{C.F.} + \text{P.I.}$ or $z = \phi_1(y + x) + \phi_2(y + 2x) + (1/12) \times e^{2x-y} - x e^{x+y} - (1/3) \times \cos(x + 2y)$.(b) This problem is same as part (a) except that the term e^{x+y} is missing on R.H.S. So, now you need not compute P.I. corresponding to e^{x+y} . Therefore the solution will take the form

$$y = \phi_1(y + x) + \phi_2(y + 2x) + (1/12) \times e^{2x-y} - (1/3) \times \cos(x + 2y)$$

(c) Try yourself. **Ans.** $z = \phi_1(y + x) + x\phi_2(y + x) + \phi_3(y + 2x) + x e^{y+x} - (x^2/3) \times (y + x)^{3/2}$.**Ex. 10.** Solve (i) $(D^2 - 6DD' + 9D'^2)z = \tan(y + 3x)$ [Delhi 2007; Ravishankar 2004](ii) $(D^2 - 6DD' + 9D'^2)z = 6x + 2y$

4.14 Ordinary and Partial Differential Equations

Sol (i) Here auxiliary equation is $(m-3)^2 = 0$ so that $m = 3, 3$.

$\therefore$ C.F. = $\phi_1(y+3x) + x\phi_2(y+3x)$, ϕ_1, ϕ_2 being arbitrary functions

$$\text{P.I.} = \frac{1}{(D-3D')^2} \tan(y+3x) = \frac{x^2}{1^2 \times 2!} \tan(y+3x) = \frac{x^2}{2} \tan(y+3x)$$

$\therefore$ The required solution is $z = \phi_1(y+3x) + x\phi_2(y+3x) + (x^2/2) \times \tan(y+3x)$.

(ii) Try yourself. **Ans.** $z = \phi_1(y+3x) + x\phi_2(y+3x) + 3x^3 + x^2y$.

EXERCISE 4(A)

Solve the following partial differential equations:

1. $(D^2 - DD' - 6D'^2)z = \cos(2x+y)$ [Agra 2009, 10]

Ans. $z = \phi_1(y+3x) + \phi_2(y-2x) - (1/4) \times \cos(2x+y)$, ϕ_1, ϕ_2 being arbitrary functions.

2. $r - 4s + 4t = e^{2xy}$ [Agra 2010]

Ans. $z = \phi_1(y+2x) + x\phi_2(y+2x) + (x^2/2) \times e^{2xy}$, ϕ_1, ϕ_2 being arbitrary functions

3. $(D^3 - 4D^2D' + 4DD'^2)z = 6 \sin(3x+2y)$

Ans. $z = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x) + 2 \cos(3x+2y)$, ϕ_1, ϕ_2, ϕ_3 being arbitrary functions.

4. $(D-3D')(D+3D')z = e^{3xy}$ [Agra 2005]

Ans. $z = \phi_1(y+3x) + x\phi_2(y+3x) + \phi_3(y-3x) + (x/12) \times e^{3xy}$, ϕ_1, ϕ_2, ϕ_3 being arbitrary functions.

4.10 Short Method II. When $f(x, y)$ is of the form $x^m y^n$ or a rational integral algebraic function of x and y .

Then the particular integral (P.I.) is evaluated by expanding the symbolic function $1/f(D, D')$ in an infinite series of ascending powers of D or D' . In solved examples 1 and 2 of Art. 4.11, we have shown that P.I. obtained on expanding $1/f(D, D')$ in ascending powers of D is different from that obtained on expanding $1/f(D, D')$ in ascending powers of D' . Since to get the required general solution of given differential equation any P.I. is required, any of the two methods can be used. The difference in the two answers of P.I. is not material as it can be incorporated in the arbitrary functions occurring in C.F. of that given differential equation.

Remark: If $n < m$, $1/f(D, D')$ should be expanded in powers of D'/D whereas if $m < n$, $1/f(D, D')$ should be expanded in powers of D/D' .

4.11 Solved examples based on short method II

Ex. 1. Solve $(D^2 - a^2 D'^2)z = x$ or $(\partial^2 z / \partial x^2) - a^2 (\partial^2 z / \partial y^2) = x$.

Sol. Here auxiliary equation is $m^2 - a^2 = 0$ so that $m = a, -a$.
 $\therefore$ C.F. = $\phi_1(y+ax) + \phi_2(y-ax)$, ϕ_1, ϕ_2 being arbitrary functions. ... (1)

$$\begin{aligned} \text{Now, P.I.} &= \frac{1}{D^2 - a^2 D'^2} x = \frac{1}{D^2 [1 - a^2 (D'^2 / D^2)]} x = \frac{1}{D^2} \left(1 - a^2 \frac{D'^2}{D^2} \right)^{-1} x \\ &= \frac{1}{D^2} \left(1 + a^2 \frac{D'^2}{D^2} + \dots \right) x = \frac{1}{D^2} x + \frac{x^3}{6}. \end{aligned} \quad \dots (2)$$

Alternatively, we can compute P.I. on follows :

$$\text{P.I.} = \frac{1}{D^2 - a^2 D'^2} x = \frac{1}{-a^2 D'^2 [1 - (D^2 / a^2 D'^2)]} x = -\frac{1}{a^2 D'^2} \left(1 - \frac{D^2}{a^2 D'^2} \right)^{-1} x$$